



Hands-on Principal Architect · 19+ yrs · Python (FastAPI, LangChain, LangGraph, MCP) · RAG + LLM in production · AWS · Java + Spring Boot · GNN · HIPAA, ISO 27001 · AI-Leveraged Delivery (Claude Code + OpenAI Codex)

SUMMARY

Hands-on principal architect, 19+ years, current stack **Python** — **FastAPI** web application programming interfaces (APIs), **LangChain**, **LangGraph** and **Model Context Protocol (MCP)** servers behind **ten** production analytics and artificial intelligence (AI) services for a chronic-care platform on Amazon Web Services (AWS). **Claude Code** and **OpenAI Codex** in daily use across a **20-engineer** team — a measured rollout with hands-on review as the guardrail. Retrieval-augmented generation (RAG) runs on **Elasticsearch dense_vector** across **80+ chronic conditions**; self-hosted embeddings and **Llama 3.1** keep protected health information inside the client boundary; large language model (LLM) guardrails are designed to reject uncited claims. That platform runs on **Docker**, **Kubernetes** and **Terraform**, holding deploys under **10 minutes** with infrastructure cost down **25–30%**. Behind it, **11 years of Java and Spring Boot across three eras** — multi-tenant software as a service (SaaS) for regulated banking customers and central single sign-on (SSO) with **zero user disruption**.

Key Skills

Technology Leadership & Strategy

- **Platform Architecture Ownership** (Target-State Design, Technology Selection, Migration Path)
- **Engineering Standards & Design Review** (Quality Gates, Adoption Across Teams)
- **AI-Assisted Development Rollout** (Tooling, Review Practice, Safety Guardrails, Team Adoption)
- **Technology Strategy & Architecture**
- **Decision-Intelligence Platforms** (Event-Sourced + Graph + AI)
- **Cross-Functional Leadership & Stakeholder Alignment**

Engineering Excellence

- **Python Services in Production** (FastAPI, Django, Pydantic v2, Asynchronous Web APIs)
- **Agentic & Large Language Model Orchestration** (LangChain, LangGraph, Model Context Protocol Servers)
- **Retrieval-Augmented Generation** (Elasticsearch dense_vector, OpenSearch, Filtered Nearest-Neighbour Search, Embeddings)
- **LLM Productionisation** (Prompt Design, Guardrails, Field-Level Claim Grammar, Replay Determinism)
- **Retrieval & Response Evaluation** (Golden Fixtures, Normalised Discounted Cumulative Gain, Warm-Path Latency Budgets)
- **Self-Hosted Models Inside the Virtual Private Cloud (VPC)** (Ollama, Llama 3.1, Amazon SageMaker Embeddings, Self-Hosted LangSmith)
- **AI-Assisted Engineering at Team Scale** (Claude Code + OpenAI Codex across 20 Engineers, Review as the Guardrail)
- **Shared Platform Application Programming Interfaces (APIs) & Libraries** (Contracts, API Versioning, Backward Compatibility, 20+ Enterprise Consumers)
- **Vector, Relational & NoSQL Data Layers** (Elasticsearch dense_vector, OpenSearch, PostgreSQL, DynamoDB, Redis, Amazon Neptune + Gremlin)
- **Cloud Infrastructure & Platform Engineering** (AWS — Lambda, Elastic Container Service Fargate, SageMaker, Aurora, DynamoDB, Kinesis)
- **Containers & Infrastructure-as-Code** (Docker, Kubernetes, Terraform)
- **Identity & Access** (OAuth 2.0, OpenID Connect, JSON Web Tokens (JWT), Keycloak to Red Hat single sign-on, Role-Based Access Control (RBAC))
- **Multi-Tenant SaaS Architecture** (Shared-Schema, Per-Tenant Isolation, Regulated Carve-Outs)
- **Java + Spring Boot Microservices** (Java 8/17, Spring Boot 2.x, JSP — 11 years across three eras)
- **Applied Machine Learning in Production** (Graph Neural Networks (GNN) on Amazon Neptune, Leakage-Safe Featurisation, Propensity Matching)
- **Testing & Quality Gates** (Golden-Fixture Regression, End-to-End Automation, k6 Load Testing, Design-Doc Review)
- **DevOps & Continuous Integration / Continuous Delivery (CI/CD)** (Jenkins to GitHub Actions, ISO 27001)
- **Observability & Reliability Engineering** (Datadog, Amplitude, ELK — Elasticsearch, Logstash, Kibana)

Business & Delivery

- **Security, Compliance & Data Privacy** (Health Insurance Portability and Accountability Act (HIPAA) Pipelines in an ISO 27001-Certified Environment, Least-Privilege)
- **Regulated Business-to-Business (B2B) Domains** (Healthcare, Regulated Banking Customers, Web Content Accessibility Guidelines (WCAG) 2.2 Level AA)
- **Distributed United States–India Delivery Teams**
- **Enterprise Onboarding & Consolidation at Multi-Hundred-Instance Scale**
- **Performance Optimization & Cost Management**
- **Agile / Scrum Delivery Management**

PROFESSIONAL EXPERIENCE

Principal Software Architect	Mar '25 - Apr '26
Voltuswave Technologies India Pvt Ltd	Hyderabad
• Assembled a 10-query decision-intelligence platform in Python — FastAPI, LangChain, LangGraph — on 4 reusable archetypes carrying all ten production services • Drove a measured Claude Code and OpenAI Codex rollout across a 20-engineer team, covering daily design exploration, first-cut implementation and test scaffolding under review	

- Engineered a retrieval-augmented generation (RAG) pipeline on **Elasticsearch dense_vector** filtered nearest-neighbour search across **80+ chronic conditions**, keeping **Amazon Neptune** off the serving path
- Enforced **large language model (LLM)** narrative guardrails — pinned model, temperature 0, chain hash, field-level claim grammar — designed to reject **100% of uncited claims**
- Secured protected health information inside the client **virtual private cloud** by self-hosting **Amazon SageMaker** embeddings and **Llama 3.1** on **Ollama** for every model call
- Reimagined the data layer across **DynamoDB**, **PostgreSQL** and **OpenSearch** with partition-key and indexing design holding **P95 under 30 ms** at 100,000-user load
- Provisioned **Amazon Web Services** with **Docker**, **Kubernetes**, **Terraform** and **Elastic Container Service Fargate at 70% Spot** — deploys **under 10 minutes**, infrastructure cost down **25–30%**

Lead SDE/Sr. Staff Engineer / Staff Software Engineer Dec '19 - Feb '25

Deque Software Hyderabad

- Forged a central authentication service on **OpenID Connect** and **OAuth 2.0**, migrating every instance from **Keycloak** to **Red Hat single sign-on** with zero disruption
- Productized a billing and subscription service consumed by **every product**, porting it from **Node.js/TypeScript** to **Java and Spring Boot** and gating feature access by subscription
- Exposed shared platform APIs to **20+ enterprise clients** — Jira integration, API versioning, customer-specific frontends on one **Spring Boot** backend, strict backward compatibility
- Implemented multi-tenant isolation across **approximately 300 enterprise instances**, adding an organisation ID to every table and resolving tenant from the **JWT**
- Steered **70–80%** of customers onto shared **axe Monitor** instances on **Spring Boot 2.x and Java 8/17**, one codebase also serving regulated banks on dedicated databases
- Stewarded a **zero-downtime** upgrade from **PostgreSQL 9.6** to **Amazon Aurora PostgreSQL 13.x** across every axe Auditor instance
- Modernised **CI/CD** from **Jenkins** to **GitHub Actions**, achieving **ISO 27001** and **50% faster** release cycles
- Slashed regression testing time **over 50%** with an end-to-end automated testing framework built on **Puppeteer**
- Championed decomposition of a monolith into **10–15 services**, splitting on independent scaling and database ownership rather than on team boundaries

Software Engineer III Aug '18 - Jul '19

Rocket Software Bangalore

- Anchored decomposition of **IBM Db2 Configuration Manager** into **Java and Spring Boot** microservices, defining boundaries across **z/OS and Linux, UNIX and Windows** platforms
- Coached a **5-engineer** team through delivery planning, code review and performance appraisals while remaining hands-on as an individual contributor
- Surfaced defect hot-spots through **REST** interfaces and service isolation, reducing reported bugs **over 50%**
- Partnered with product to extend platform support to **Windows** and Linux — Ubuntu, CentOS and **Red Hat Enterprise Linux (RHEL)** — raising sales **over 20%**

Co-Founder and VP of Technology Mar '17 - Apr '18

Voltuswave Technologies India Pvt Ltd Hyderabad

- Co-founded VoltusWave and launched its first **application Platform-as-a-Service (aPaaS)** product for web and Android, winning **8 early enterprise clients**
- Mobilized an engineering organisation from **0 to 35** in 13 months, mentoring freshers and senior engineers while establishing delivery process
- Helmed development, deployment and operations on **Amazon Elastic Container Service**, setting the architecture roadmap every subsequent product inherited

Senior Software Engineer Mar '16 - Mar '17

Teletext India Private Limited, Hyderabad Hyderabad

- Codified a scatter-gather integration across **25 hotel suppliers** on a **MuleSoft enterprise service bus**, succeeding where a prior **.NET 2.0** upgrade had failed
- Cut operational cost **46%** by retiring a third-party pricing product for an in-house **ELK** cache with a 20-minute time-to-live
- Reconciled supplier hotel identifiers onto **GIATA MultiCodes**, automating the pipeline on **AWS Lambda** with human approval retained, growing revenue **8% month over month**
- Lifted hotel booking conversions **65%** by widening matched inventory and ranking contracted properties first across **2–3 million daily requests**

Technical Architect May '15 - Nov '15

CURA Software Solutions Hyderabad

- Introduced **Linux/Unix deployment** for the Governance, Risk & Compliance platform, cutting running costs and winning **12 new customers within 2 months**
- Ported the data model from **Microsoft SQL Server** to **MySQL**, enabling the low-cost Linux deployment

Technical Lead Jul '13 - May '15

innRoad India Hotel Software Pvt. Ltd. Hyderabad

- Converted a **legacy ASP.NET 2.0** application into a modern **single-page application** using Knockout.js and a Boilerplate JavaScript framework
- Productized **REST-based WCF** services that enabled and monetised customer integrations
- Codified engineering standards and mentored the team through a full **product-modernisation effort**

Software Architect Jun '12 - May '13

McDonald's Corporation OakBrook, IL

- Modeled the schema and governance framework for an enterprise Master Data Management (MDM) program
- Forged interactive user-interface screens with drill-down, dynamic drop-downs, and full row-level editing
- Partnered with data owners to deliver a robust enterprise data-governance system across the organisation

May '12

Senior Java Developer	Apr '08 -
El Paso Corporation	Houston, TX
- Rebuilt a legacy Citrix-hosted Java application into a modern .NET 2.0 stack on SQL Server 2005 - Steered a scalable, maintainable .NET platform that replaced the outdated system while preserving core business functions - Anchored large-scale enterprise modernisation for a high-stakes energy-sector client	
.NET Developer	Apr '07 - Apr '08
LyntonWeb	Houston, TX
Developed Paymaster, a PayPal-style payments application, for a Mexican financial institution	

EDUCATION

PG Diploma in Artificial Intelligence and Machine Learning	Jan '21
IIIT Bangalore	
Master's in Computer Science	Aug '04 - Dec '07
University of Houston	Houston, Texas

CERTIFICATION

PG Diploma in Artificial Intelligence and Machine Learning, IIIT Bangalore